KING SALMAN INTERNATIONAL UNIVERSITY

Field of ENGINEERING

Engineering Program
All Engineering Program

MAT121 Statics

3 Cr. Hrs. = (***2*** LCT + ***2*** TUT + ***0*** LAB + ***0*** OTH)

Lecture : (EQUIVALENT FORCE SYSTEMS)

Prof. Ayman Abdel hamed
Dr : Haytham Essam

Date : 14 / 10 / 2026

Equivalent Force Systems

Treatment of a body as a single particle is not always possible. In general, the size of the body and the specific points of application of the forces must be considered.

Current chapter describes

1- The effect of forces exerted on a rigid body,

2- How to replace a given system of forces with a simpler equivalent system.

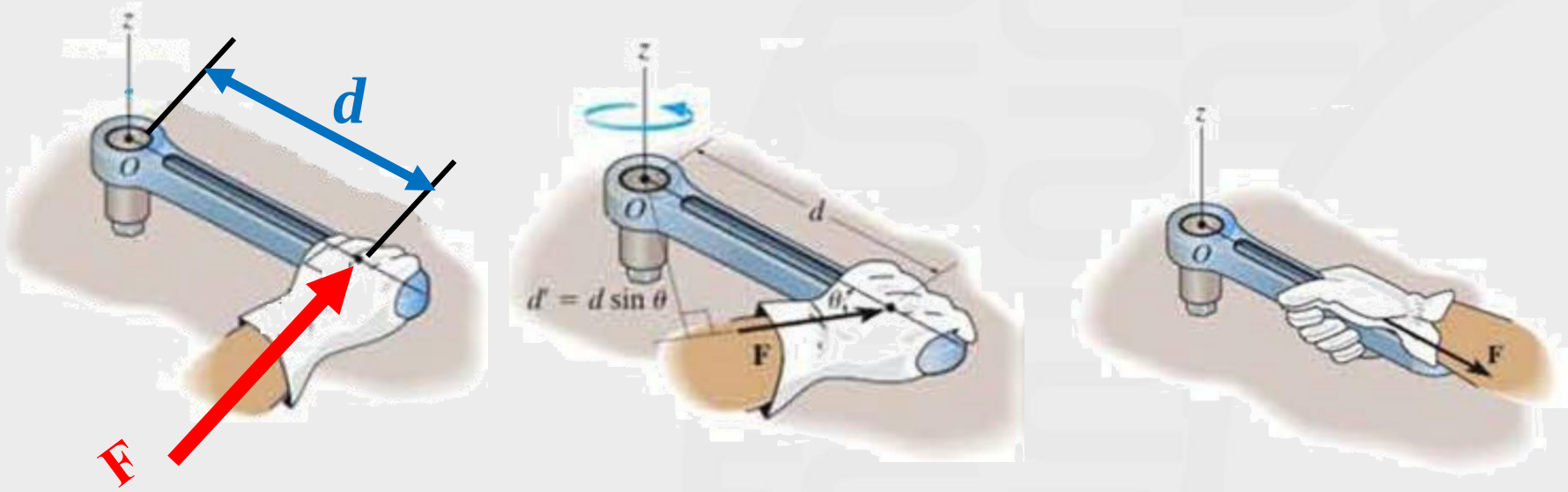
First, we need to learn some new statics concepts, including:

- Moment of a force about a point.
- Moment due to a couple.

- Any system of forces acting on a rigid body can be replaced by an equivalent system consisting of one force acting at a given point and one couple.

Moment of a Force

The effect of a force with respect to any point on its line of action is translation, and the effect of a force with respect to any point not on its line of action is rotation. This rotational effect is termed the moment of the force.



Scalar Form

Magnitude:

The magnitude of the moment can be calculated :

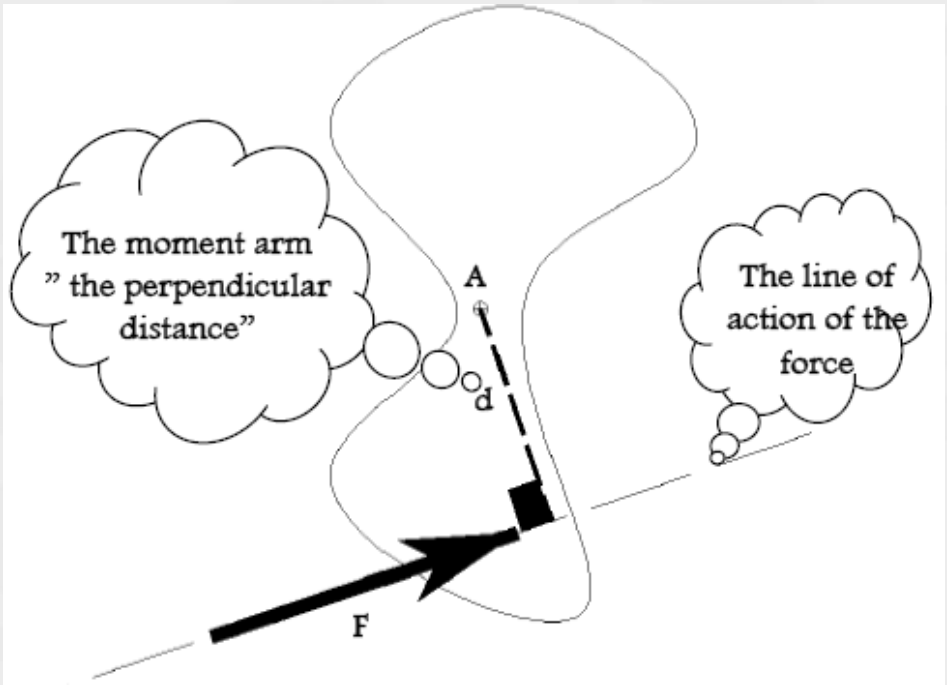
$M_A = F * d$

Where:

M_A is a moment about point A

F is the applied force

d is the moment arm or the perpendicular distance from point A to any point on the line of action of the force.



Direction:

The direction of the moment is perpendicular to the plane of the force and the moment arm.

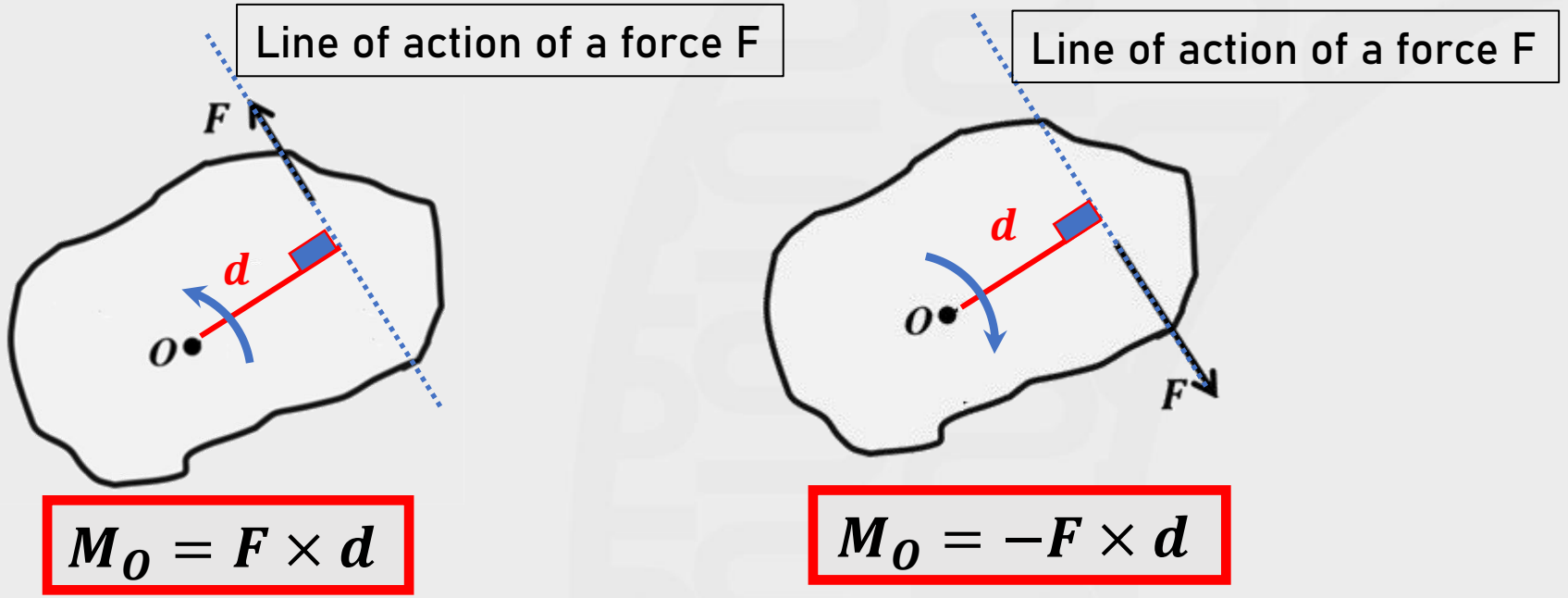
Units of the moment

Newton. meter (N. m).

Moment of a force (2D)

Consider, a rigid slab acted upon by a force $\boldsymbol{F}$ as shown in Figure

The moment of $\boldsymbol{F}$ about a point O chosen in the plane of the Figure is represented by a vector $\boldsymbol{M_o}$ perpendicular to that plane :

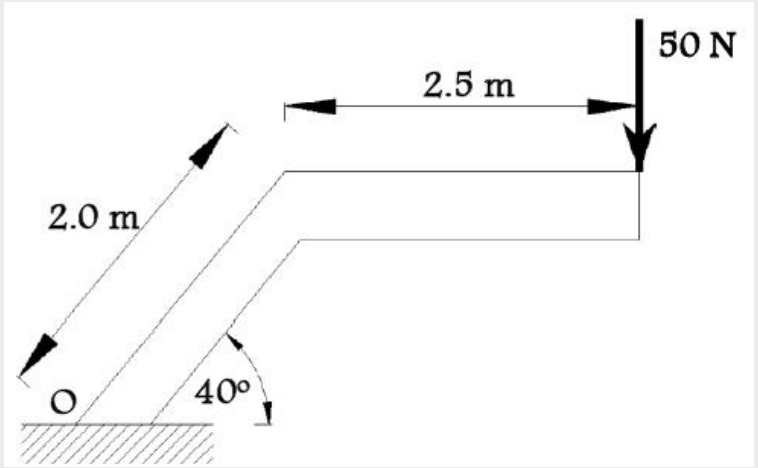


$$\boldsymbol{M_O = F \times d}$$

$$\boldsymbol{M_O = -F \times d}$$

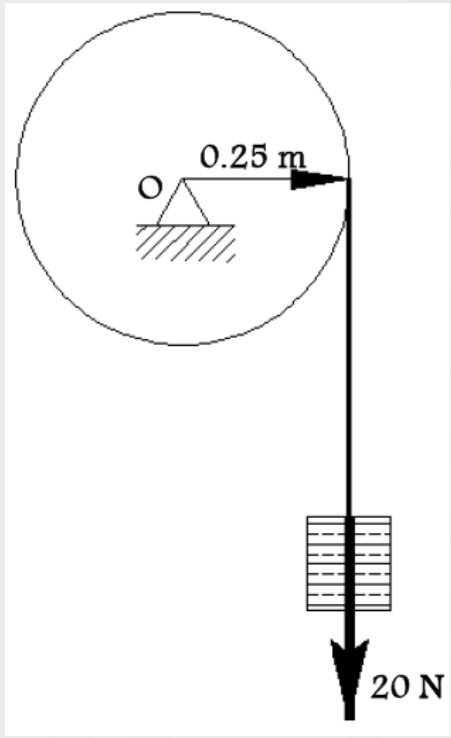
Example 2 – 1:

For the shown figures, determine the moment of the force about point "O"



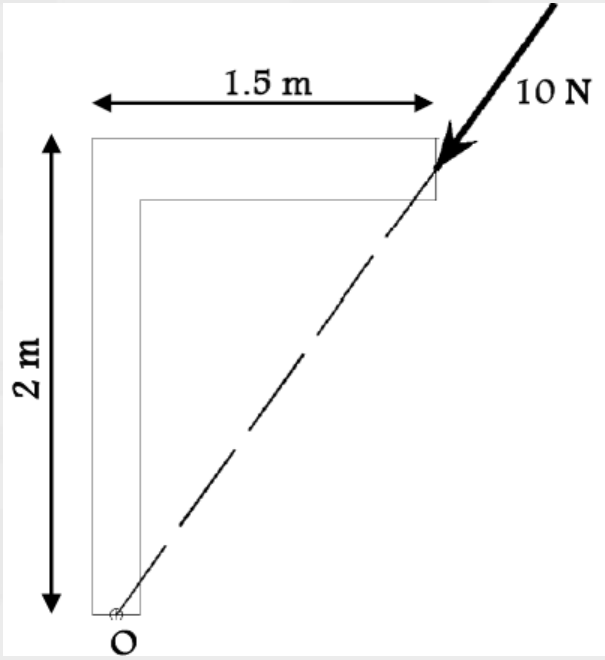
$$M_O = F \times d$$

$$M_O = 50 \times (2.5 + 2\cos 40) = 201.6\ N.m \curvearrowright$$



$$M_O = F \times d$$

$$M_O = 20 \times 0.25 = 5\ N.m \curvearrowright$$

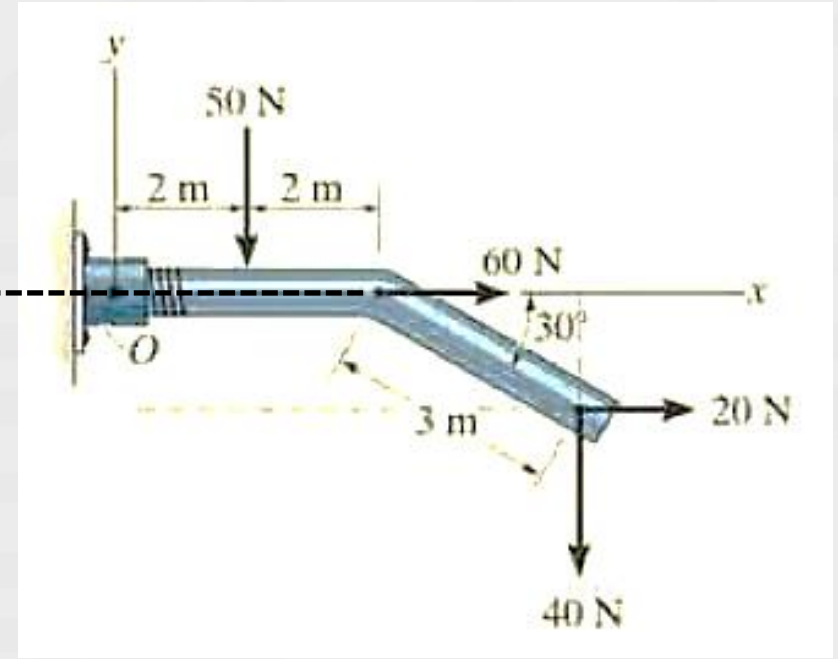


$$M_O = F \times d$$

$$M_O = 10 \times Zero = Zero\ N.m$$

Example 2-2

Determine the resultant moment of the four forces acting on the rod shown in Figure about point O



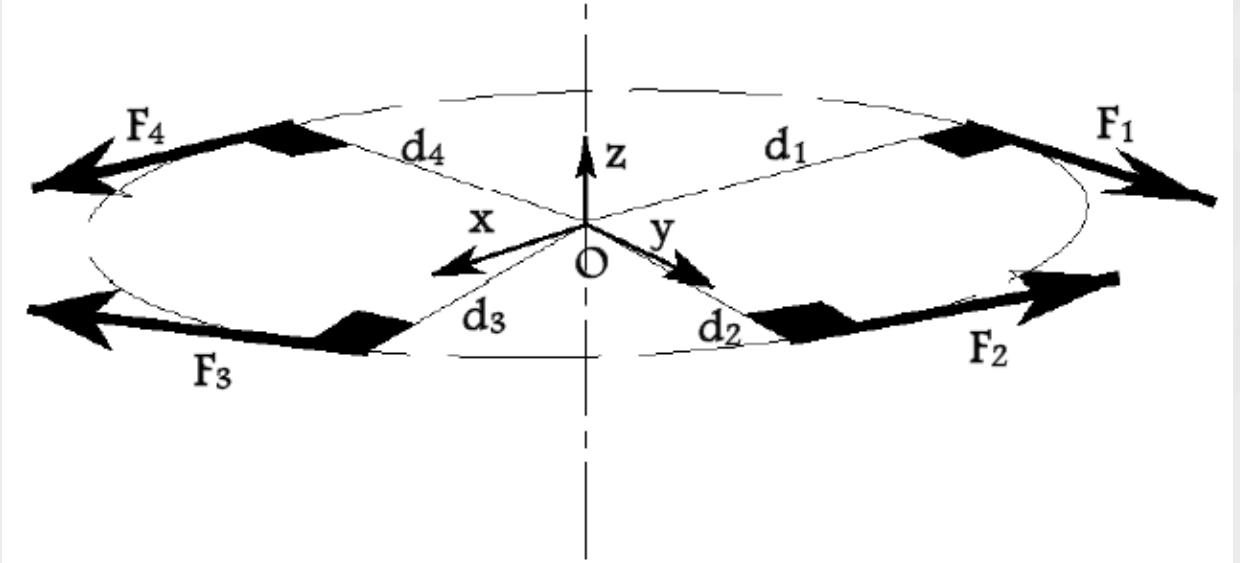
SOLUTION:

$$M_O = -(50 \times 2) + (60 \times 0) + (20 \times 3sin\,30) - (40 \times (4 + 3\cos 30))$$

$$M_O = -334\ N.m$$

Resultant Moment of a System of Coplanar Forces:

For two dimensional problems where all the forces lie within the x – y plane then the direction of the moments will be in z direction as shown in figure, so the resultant moment M_O about point O can be determined by finding the algebraic sum of the moments caused by all the forces in the system.



Resultant Moment of Coplanar Forces

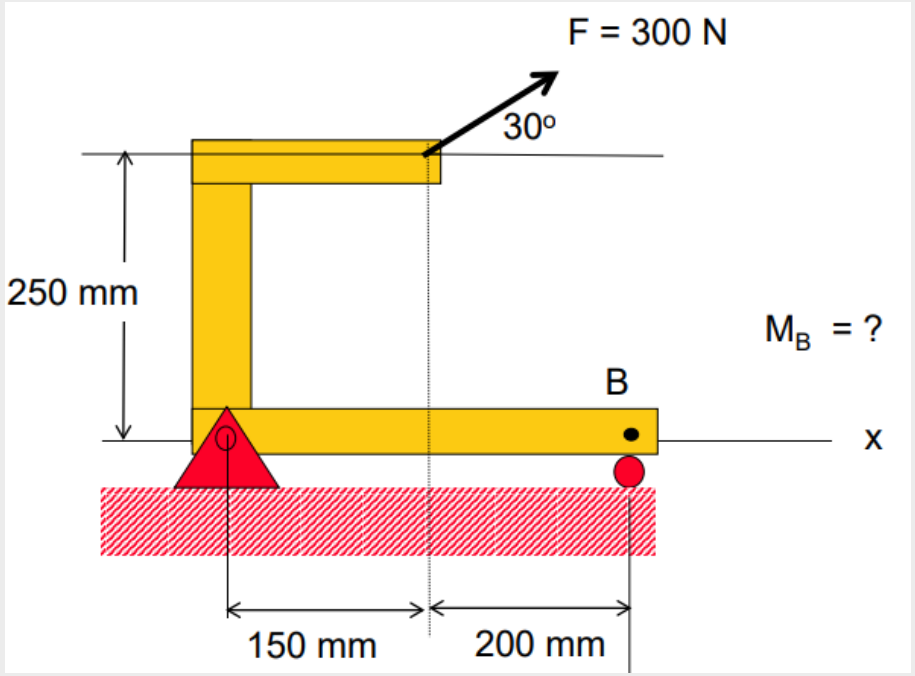
$$\circlearrowleft_+ M_O = \sum_{i=1}^{N} (F_i \times d_i)$$

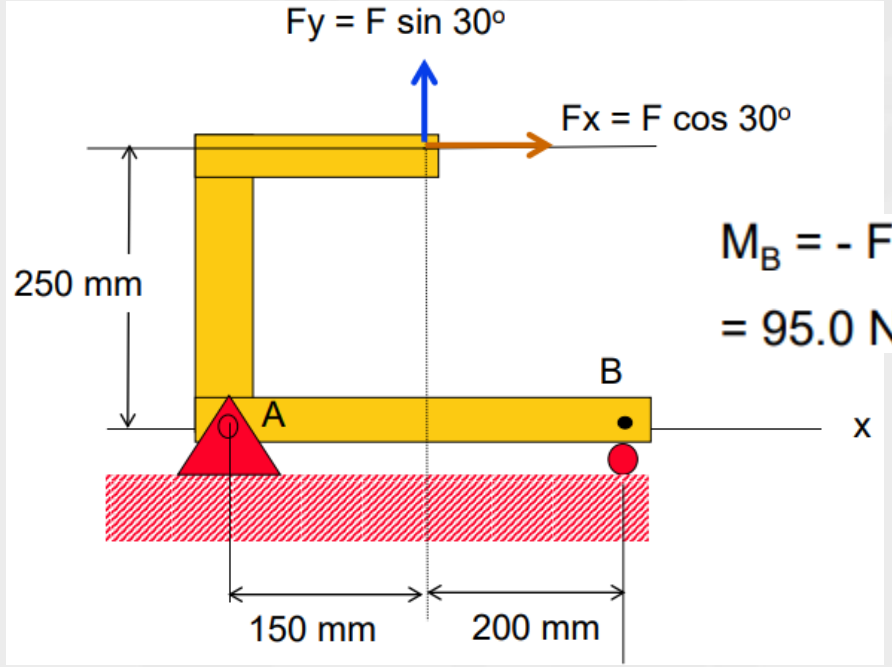
$$M_O = -F_1 \times d_1 + F_2 \times d_2 - F_3 \times d_3 + F_4 \times d_4$$

Varignon's Theorem:

Varignon's Theorem, also often called the **principle of moments**, is a very useful tool in scalar moment calculations. In cases where the perpendicular distance is hard to determine, Varignon's Theorem offers an alternative to finding that distance.

> *" The moment of a force about a point is equal to the sum of the moments of the force's components about the same point ".*





M_B = - Fx 0.250 – Fy 0.2 = -95.0 N.m

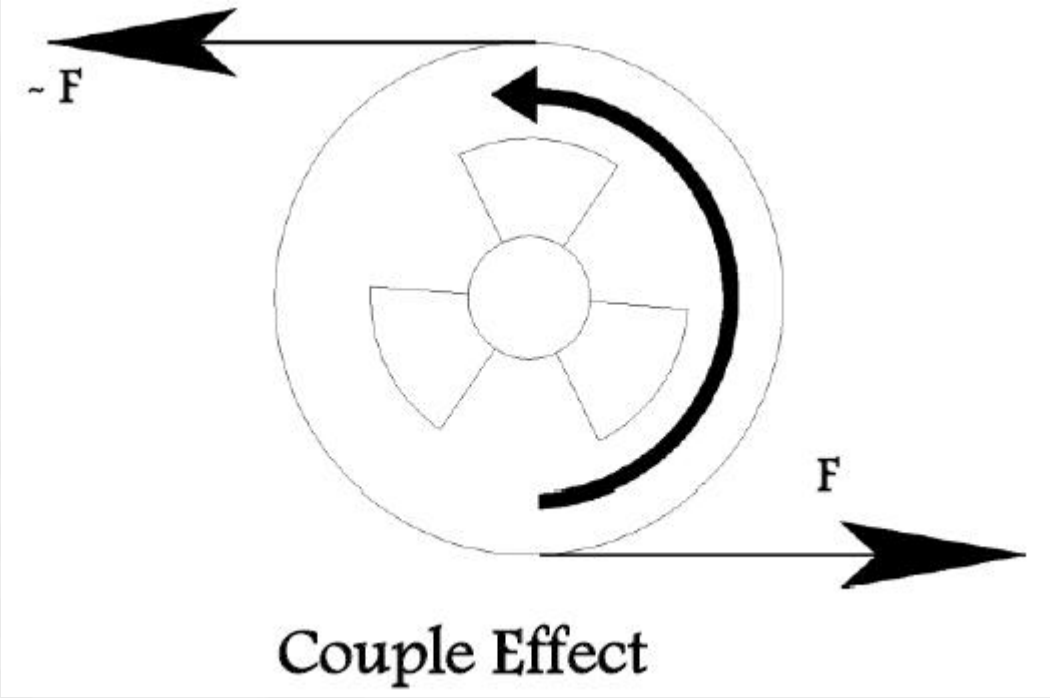
= 95.0 N.m counterclockwise

Moment of a Couple

A couple is a system of :

1. Same in magnitude
2. parallel forces acting in opposite direction
3. Not in the same straight line, as shown in figure

- The moment of a couple about any point constant.



Couple Effect

1- Scalar Form

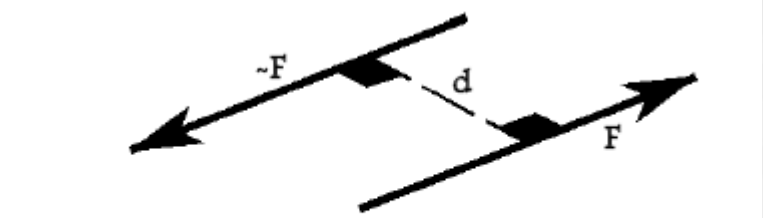
Magnitude:

The magnitude of the moment can be calculated as:

$M = F * d$

Where:

d is a perpendicular distance between the two forces of the couple, and it is also called (couple arm).



Couple Moment Arm for Scalar Form

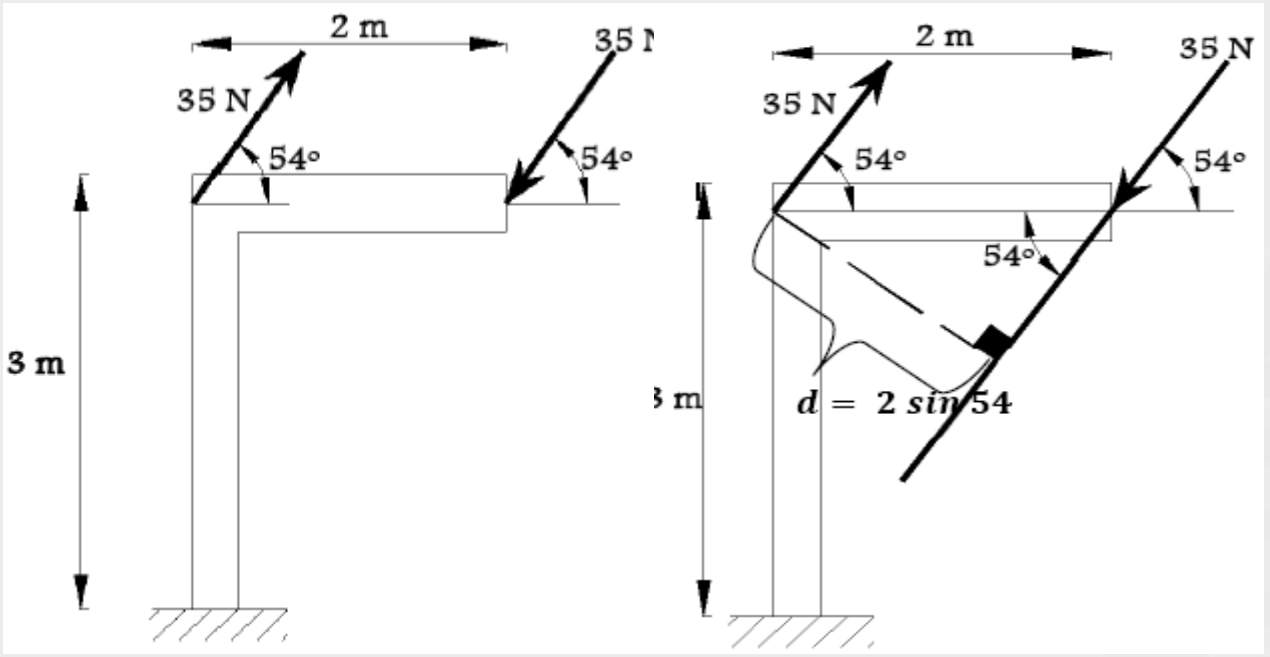
Direction:

The direction of the moment is perpendicular to the plane of the forces and arm.

- The sense of the moment can be determined according to right hand rule as explained before.
- The couple moment does not depend on the position of the point at which the moment is calculated but only depend on the distance between the two forces of the couple

Example 2 – 3:

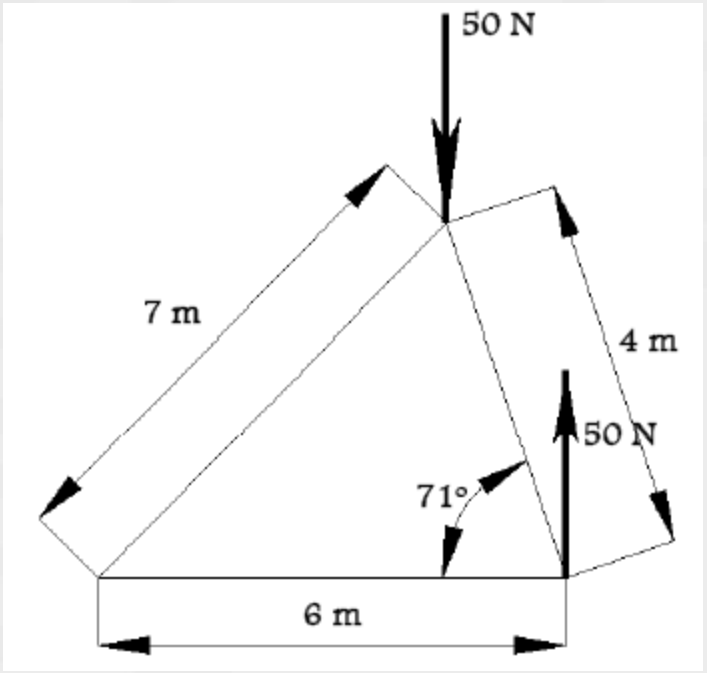
Determine the magnitude of the couple moment for the shown couples of the following figures.



$\because M = F \times d$

$\therefore M = 35 \times 2 \sin 54$

$= 56.631\ N.m$



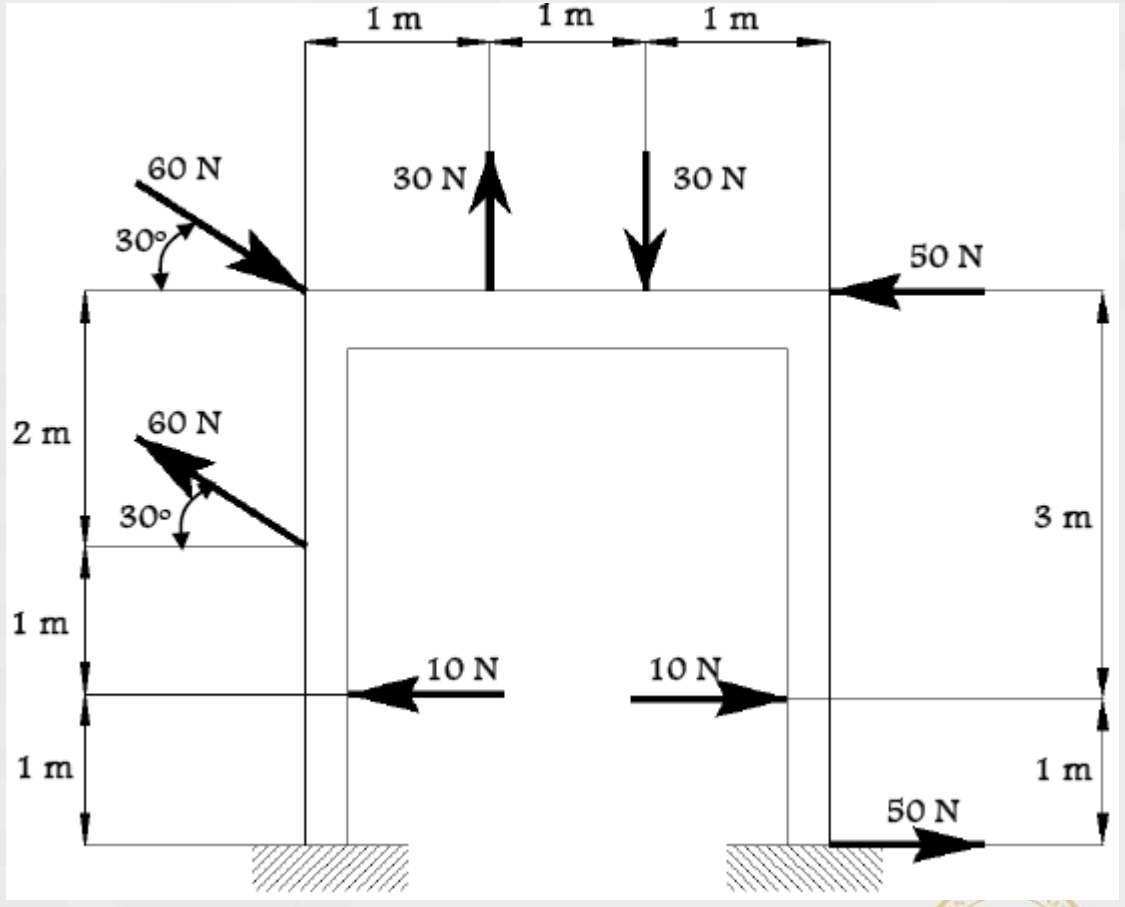
$\because M = F \times d$

$\therefore M = 50 \times 4 \cos 71$

$= 65.113\ N.m$

Example 2 – 4:

For the shown figure, determine the magnitude of the resultant moment acting on the shown body.



- For couple (50, - 50):

$$\because M = F \times d$$

$$\therefore M_1 = 50 \times 4 = 200\ N.m$$

- For couple (30, – 30):

$$M_2 = -30 \times 1 = -30\ N.m$$

- For couple (60, – 60):

$$M_3 = -\ 60 \cos 30 \times 2 = -103.923\ N.m$$

- For couple (10, – 10):

$$M_4 = 10 \times Zero = Zero$$

$$\curvearrowleft_{+} \therefore \boldsymbol{M_R = \sum_{i=1}^{N} M_i = 200 - 30 - 103.923 = 66.077\ N.m}$$

KING SALMAN INTERNATIONAL UNIVERSITY

THANK YOU

16949 www.ksiu.edu.eg info@ksiu.edu.eg

El Tur Campus : Sahely Road Ras suder Campus : Obour Street Sharm El Sheikh Campus : El Nour District